Highlights of the Compendium of Philippine Environment Statistics 2016-2025 Component 3: Residuals

HIGHLIGHTS OF COMPONENT THREE

RESIDUALS

The Component 3: Residuals of the Compendium of Philippine Environment Statistics compiles statistics on the amount and characteristics of residuals generated by human production and consumption processes, their management, and their final release to the environment. Residuals are defined as flows of solid, liquid, and gaseous materials, and energy that are discarded, discharged, or emitted by establishments and households through processes of consumption, production, or accumulation.

Residuals, including their volume, type, source, location and trends, can significantly impact both humans and the environment. Compiling these statistics is essential for evidence-based policy-making, especially in developing environmental regulations aimed at mitigating these impacts. In addition, the statistics in this component are useful in developing various environmental accounts, such as water and material flow accounts, as they form a key part of supply and use accounts.

The statistics under this component are connected to the driver, adaptation and mitigation thematic areas of the Global Set of Climate Change Statistics and Indicators. Moreover, statistics compiled under this component may help provide indicators to achieve the following Sustainable Development Goals: Goal 2 – End hunger, achieve food security and improved nutrition, and promote sustainable agriculture; Goal 6 – Ensure availability and sustainable management of water and sanitation for all; Goal 9 – Build resilient infrastructure, promote inclusive and sustainable industrialization, and foster innovation; Goal 11 – Make cities and human settlements inclusive, safe, resilient and sustainable; Goal 12 – Ensure sustainable consumption and production patterns; and Goal 13 – Take urgent action to combat climate change and its impacts.

3.1 Emissions to Air

This subcomponent covers the sources and quantities of gaseous and particulate substances released to the atmosphere by establishments and households as a result of production. Results of emissions are estimated as inventories by type of source, by process, and by economic activity. Statistics on emissions to air can be used to define the maximum allowable level of emission as well as to inform about the impacts of air pollution on humans.

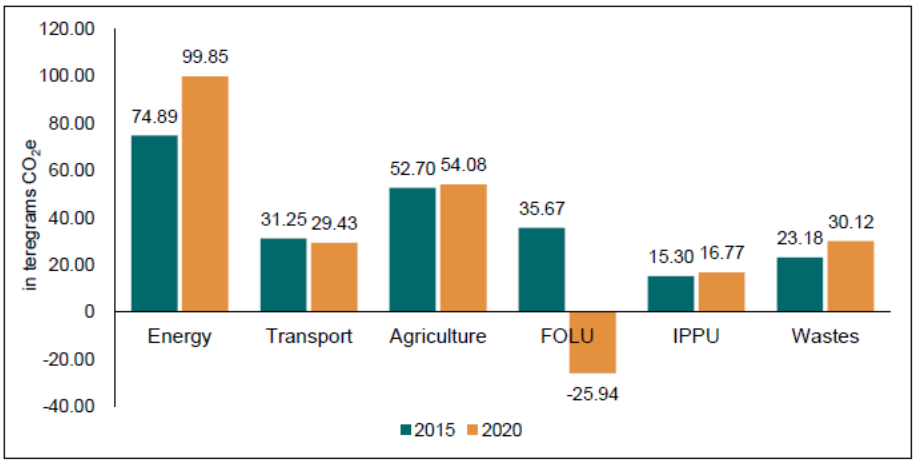
3.1.1 Emissions of Greenhouse Gases

Greenhouse gases (GHGs) are substances that absorb infrared radiation and contribute to the increase in temperature of the Earth's surface and the lower atmosphere. GHG emission inventories are compiled based on the guidelines developed by the Intergovernmental Panel on Climate Change. The GHG inventory presents the contribution of six sectors to the country's emissions and removals, namely: 1) Energy; 2) Transport; 3) Agriculture; 4) Forestry and Other Land Use (FOLU); 5) Industrial Processes and Product Use (IPPU); and 6) Waste.

Based on the report of the Climate Change Commission, the country's total GHG emissions amounted to 204.33 teragrams of carbon dioxide equivalent (Tg CO_2e) in 2020, a decline of 12.3 percent from 232.99 Tg CO_2e in 2015. The energy, trar agriculture, IPPU and waste sectors collectively emitted a total of 230.26 Tg CO_2e, while the FOLU sector sequesterec

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Figure 3.1. Greenhouse Gas Emissions, 2015 and 2020



Source: Climate Change Commission

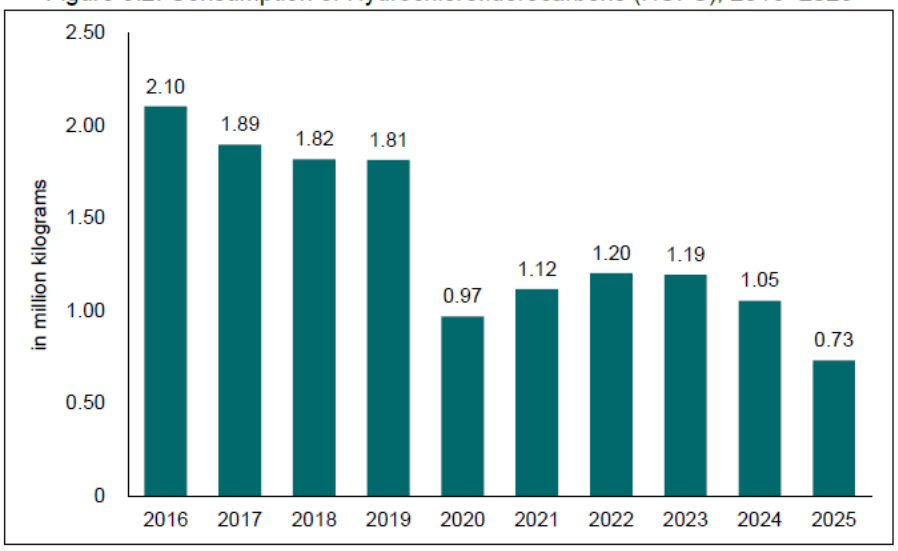
3.1.2 Consumption of Ozone Depleting Substances

Ozone Depleting Substances (ODS) are chemicals that destroy the earth's protective ozone layer. The consumption of ODS covers statistics on chlorofluorocarbons (CFCs), hydrochlorofluorocarbons (HCFCs), halons, methyl chloroform, carbon tetrachloride, methyl bromide, and other ODS.

The Montreal Protocol is a global agreement that commits parties to take concrete measures to protect the ozone layer by freezing, reducing, or ending production and consumption of controlled substances. ODS, which include HCFC and methyl bromide, are controlled under the Montreal Protocol and its amendments.[1]

The country's consumption of HCFC decreased by 30.6 percent, to 731.7 thousand kilograms in 2025 from 1.05 million kilograms in 2024. On the other hand, methyl bromide consumption reached to 26.11 thousand kilograms, an increase of 60.0 percent from 16.32 thousand kilograms in 2024. (Figure 3.2 and Table 3.3)

Figure 3.2. Consumption of Hydrochlorofluorocarbons (HCFC), 2016–2025



Source: Environmental Management Bureau

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3.2 Generation and Management of Wastewater

aid in the assessment and policy formulation on wastewater volume and pollution levels. The Metropolitan Waterworks and Sewerage System reports data on the amount of wastewater collected and treated prior to release to the environment.

Ideally, the coverage of reporting is at the national level. However, due to limitations on data availability, statistics under this subcomponent cover only the East and West Zones of Metro Manila.

3.2.1 Generation and Pollutant Content of Wastewater

This topic includes statistics on the volume of water that is no longer required, thus discarded by the user, and statistics on the amount of pollutants contained in wastewater before collection.

3.2.2 Collection and Treatment of Wastewater

Wastewater may either be discharged directly into the environment or may first be collected through sewerage systems for treatment at wastewater treatment plants. This topic focuses on the collection and treatment of wastewater which may be treated before discharge or directly discharged to the environment.

Furthermore, this includes statistics on the volume of wastewater collected and transported to discharge points or treatment facilities, the volume of wastewater treated by treatment type (primary, secondary, or tertiary), physical infrastructure for wastewater collection and treatment (e.g., the number and capacities of treatment plants), pollutants removed during treatment processes, and other pertinent information.

3.2.3 Discharge of Wastewater to the Environment

This topic captures information at the stage of final discharge of wastewater to the environment. This involves statistics on the total volume of wastewater discharged to the environment without treatment and after treatment. This may be by type of treatment and type of treatment facility (public, private, municipal, and industrial) and statistics on the pollutant content of discharged wastewater.

As of 2025, the East Zone of Manila operates 28 treatment plants with a combined capacity of 724.11 billion liters of wastewater per day, while the West Zone has 25 treatment plants capable of treating up to 723.52 billion liters per day. (Table 3.4)

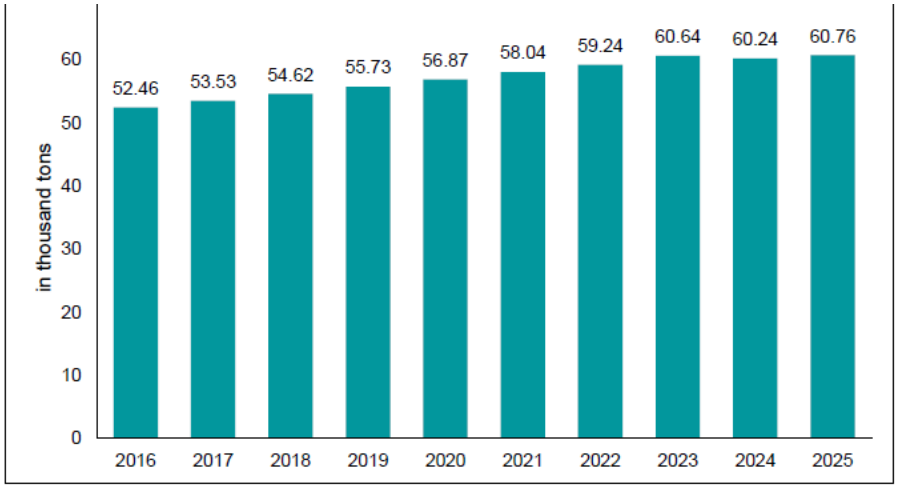
3.3 Generation and Management of Waste

This subcomponent covers statistics on the amount and characteristics of waste. Waste includes discarded materials, in solid or liquid state, that are no longer needed by the owner or user. Statistics on the generation and management of waste aid the formulation of programs that aim to reduce waste generation through prevention, reduction, recycling, and reuse.

3.3.1 Generation of Waste

Statistics under this topic describe the amount of waste generated before any collection or treatment. The broad waste categories frequently used in waste statistics are municipal, industrial, and hazardous waste. Core statistics under this topic are the amount of waste generated by source and amount of hazardous waste generated. Hazardous waste, as defined in the FDES 2013, is a special group of waste that, due to its toxic or other hazardous character, requires special management and is controlled by law in many countries. In the Philippines, the management of hazardous wastes is governed by Republic Act No. 6969, known as the "Toxic Substances and Hazardous and Nuclear Wastes Control Act of 1990". Meanwhile, the Republic Act No. 9003 or the "Ecological Solid Waste Management Act of 2000", ensures proper segregation, collection, treatment of solid waste.

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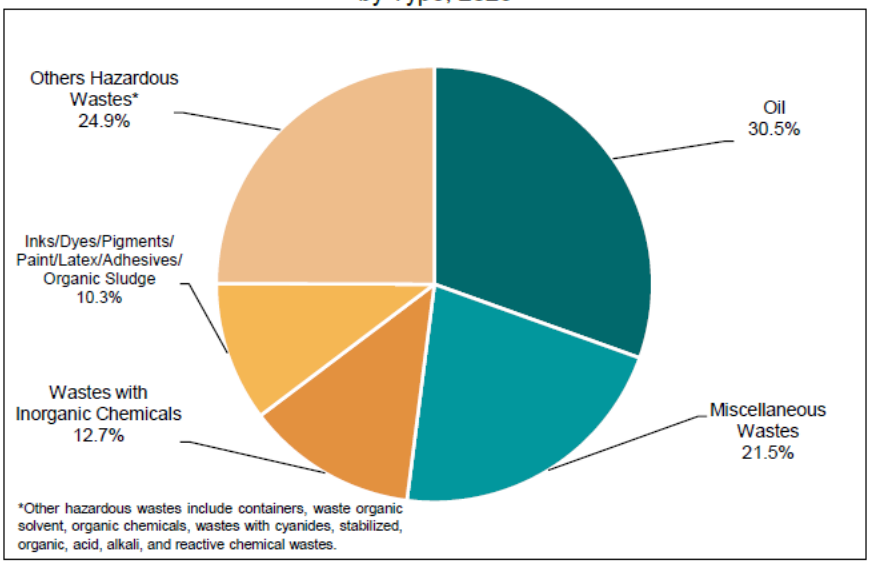
Source: Environmental Management Bureau

3.3.2 Management of Waste

This topic covers statistics on the amount of waste collected and transferred to treatment facilities or to their final disposal, the amount of waste treated and disposed by type of treatment and disposal, the number and capacity of treatment and disposal plants and other relevant information. Out of the seven core statistics, only data on hazardous waste treatment, and the number of solid waste and hazardous waste facilities are included in this publication.

Based on the report of the Environmental Management Bureau, the country generated a total of 285.41 thousand tons of hazardous waste in 2025. Among the types of hazardous waste, oil was the largest with 87.01 thousand tons or 30.5 percent of the total generated hazardous waste. This was followed by miscellaneous wastes at 61.45 thousand tons or 21.5 percent, and wastes with inorganic chemicals at 36.21 thousand tons or 12.7 percent of the total generated hazardous waste. (Figure 3.4 and Table 3.6)

Figure 3.4. Percent Distribution of Generated Hazardous Waste by Type, 2025

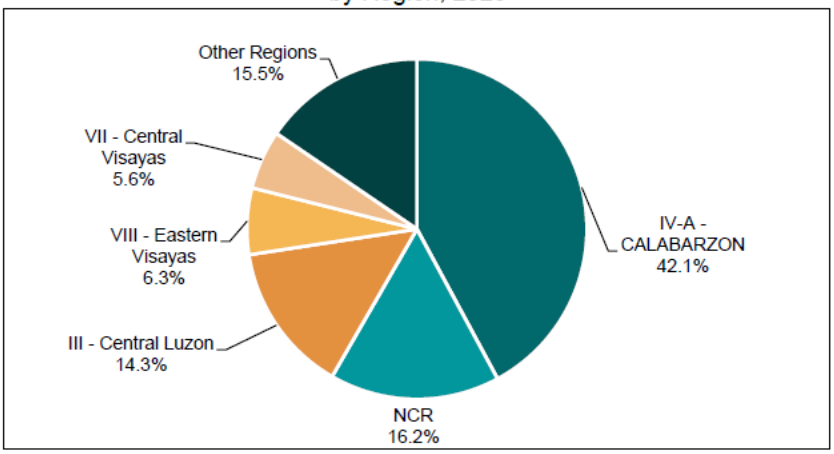


Source: Environmental Management Bureau

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The Environmental Management Bureau also reported data on the volume of treated hazardous waste. Out of the generated hazardous waste, 285.33 thousand tons were treated. Region IV-A had the largest amount of hazardous waste treated with 120.20 thousand tons or 42.1 percent of the total hazardous waste treated in the country. This was followed by National Capital Region with 46.20 thousand tons or 16.2 percent, and Region III – Central Luzon with 40.70 thousand tons or 14.3 percent of the total hazardous waste treated. (Figure 3.5 and Table 3.8)

Figure 3.5. Percent Distribution of Treated Hazardous Waste by Region, 2025



Source: Environmental Management Bureau

3.4 Release of Chemical Substances

Based on the FDES 2013, chemical fertilizers, pesticides, and other chemicals are used to increase productivity, protect plants and animals from diseases, and enhance the quality of biological products. However, environmental effects are generated by the diffusion of these chemicals through cycling systems and the build-up of contaminants. Statistics under this subcomponent include the amount of fertilizers, pesticides, and other chemical substances, which serve as the basis for estimating the chemicals that remain in the environment. The amount of fertilizers used are also included as Tier 1 statistics under Component 2: Environmental Resources and their Use.

[1] Department of Environment and Natural Resources Administrative Order (DAO) 2013-25: Revised Regulations on the Chemical Control Order for Ozone Depleting Substances (ODS)

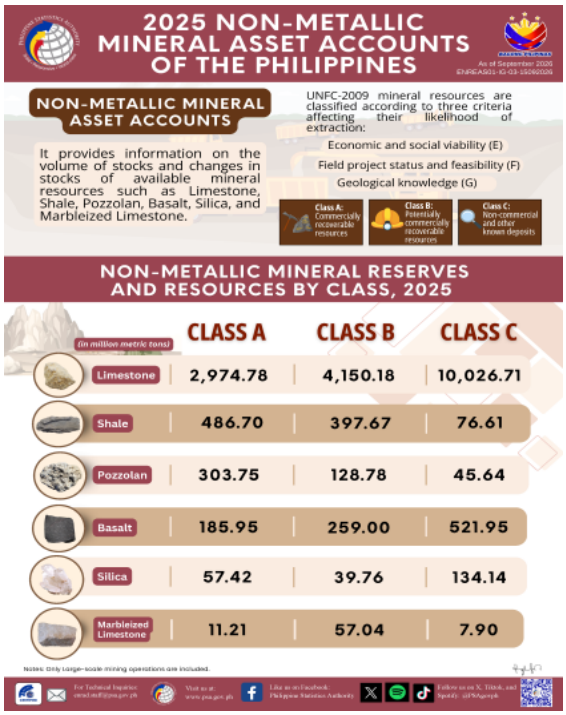
Menu

- Main Page
- Publication
  - Compendium of Philippine Environment Statistics
  - Environmental Accounts
- Environmental Accounts
  - Mineral
  - Water
  - Energy
- Environment Statistics

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Infographics



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References and Related Links

- System of Environmental-Economic Accounting (SEEA) 2012 Central Framework
- System of Environmental-Economic Accounting (SEEA) Ecosystem Accounting
- Framework for the Development of Environment Statistics (FDES) 2013

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